T1

$\xi \sim R[0, \Theta]$ ~ равн. распр.

$\underline{\tilde{\Theta}_1 = 2\bar{x}} = 2 \cdot \frac{1}{n} \cdot \sum_{i=1}^{n} x_i$

$M\xi = \int_{-\infty}^{\infty} x\, dF(x, \Theta) = \int_0^{\Theta} x \cdot \frac{1}{\Theta}\, dx = \frac{\Theta}{2}$

$M\xi^2 = \int_0^{\Theta} x^2 \cdot \frac{1}{\Theta}\, dx = \frac{\Theta^2}{3}$

$D\xi = M\xi^2 - (M\xi)^2 = \frac{\Theta^2}{3} - \frac{\Theta^2}{4} = \frac{\Theta^2}{12}$

$\forall \Theta \in [\,]$ $M\tilde{\Theta} = \Theta$ ~ несмещ.

$\forall \Theta \in [\,]$ $\forall \varepsilon > 0$ $P(|\tilde{\Theta} - \Theta| \geq \varepsilon) \xrightarrow[n \to \infty]{} 0$

① $\tilde{\Theta}_1$: $\forall \Theta > 0$ $M\tilde{\Theta}_1 = M(2 \cdot \frac{1}{n} \cdot \sum x_i) =$

$= \frac{2}{n} \sum_{i=1}^{n} Mx_i = \frac{2}{n} \cdot n \cdot M\xi = \Theta$ - несмещ

② $D\tilde{\Theta}_1 = D(2 \frac{1}{n} \cdot \sum x_i) = \frac{4}{n^2} \cdot \sum Dx_i =$

$= \frac{4}{n^2} \cdot D\xi \cdot n = \frac{\Theta^2}{3n} \xrightarrow[n \to \infty]{} 0$ - состоят по дост. усл.

$\forall \Theta > 0$

$\tilde{\Theta}_2 = x_{min}$

$\forall \Theta > 0 \quad M\tilde{\Theta}_2 = Mx_{min}$

$\xi \sim F(x)$

$\xi_{min} \sim 1-(1-F(x))^n$

$\varphi(x) = F'(x) = n(1-F(x))^{n-1} \cdot F'(x) =$

$= n\left(1-\frac{x}{\Theta}\right)^{n-1} \cdot \frac{1}{\Theta} \quad \{(0;\Theta)\}$

$Mx_{min} = \int_0^{\Theta} x \cdot n \underbrace{\left(1-\frac{x}{\Theta}\right)}_{t}^{n-1} \frac{1}{\Theta} dx =$

$t = 1-\frac{x}{\Theta}; \quad x = -(t-1)\Theta = \Theta(1-t)$

$dx = -dt$

$= -\int_1^0 n(1-t)t^{n-1}dt = \int_0^1 n(1-t)t^{n-1}\Theta dt =$

$= n\Theta\int_0^1 (t^{n-1}-t^n)dt = n\Theta\left(\frac{1}{n}-\frac{1}{n+1}\right) = \frac{\Theta}{n+1}$

смещ.

$\rightarrow \tilde{\Theta}_2' = (n+1)x_{min}$

$M\tilde{\Theta}_2' = (n+1)Mx_{min} = \Theta$ несмещ.

$D\tilde{\Theta}_2' = D((n+1)x_{min}) = (n+1)^2 \cdot Dx_{min} = \frac{n\Theta^2}{n+2} \xrightarrow[n\to\infty]{} 0$

$Mx_{min}^2 = \int_0^{\Theta} x^2 n\left(1-\frac{x}{\Theta}\right)^{n-1}dx = \int_0^1 \Theta^2(1-t)^2 n t^{n-1}dt =$

$= n\Theta^2\left(\frac{1}{n}-\frac{2}{n+1}+\frac{1}{n+2}\right) = \frac{\Theta^2 2}{(n+1)(n+2)}$

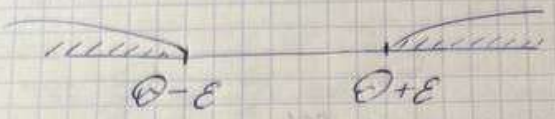
$Dx_{min} = \frac{2\Theta^2}{(n+1)(n+2)} - \frac{\Theta^2}{(n+1)^2} = \Theta^2\left(\frac{2(n+1)-(n+2)}{(n+1)^2(n+2)}\right) =$

$= \frac{n\Theta^2}{(n+1)^2\cdot(n+2)}$

не вып. дост. усл.

по определению

$$\forall \theta>0 \ \forall \varepsilon>0 \quad P(|\tilde{\theta}_2' - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$$



$$P(|\tilde{\theta}_2' - \theta| \geq \varepsilon) \geq P(\tilde{\theta}_2' \geq \theta+\varepsilon) = P((n+1)x_{min} \geq \theta+\varepsilon) =$$

$$= P\left(x_{min} \geq \frac{\theta+\varepsilon}{n+1}\right) = 1 - \Phi\left(\frac{\theta+\varepsilon}{n+1}\right) \Leftarrow$$

$$\Phi(x) = 1-(1-F(x))^n$$

$$\Leftarrow 1-\left(1-\left(1-\frac{\theta+\varepsilon}{\theta(n+1)}\right)^n\right) = \left(1-\frac{\theta+\varepsilon}{\theta(n+1)}\right)^n \xrightarrow[n\to\infty]{} e^{-\frac{\theta+\varepsilon}{\theta}} > 0$$

$\Longrightarrow$ не удовл. опред. сост.

для x_{min}:

$$P(|x_{min} - \theta| \geq \varepsilon) = P(x_{min} \leq \theta-\varepsilon) = \Phi(\theta-\varepsilon) \Leftarrow$$

ещё нет сказать

невозможное знач для x_{min} $P(x_{min} \geq \theta+\varepsilon) = 0$

$$\Leftarrow 1-(1-F(\theta-\varepsilon))^n = 1-\left(1-\frac{\theta-\varepsilon}{\theta}\right)^n =$$

$$= 1-\left(\frac{\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{\exists\varepsilon\ \exists\theta} 1 \Longrightarrow x_{min} \text{ не явл. сост.}$$

$\tilde{\Theta}_3 = x_{max}$

$M\tilde{\Theta}_3 = M x_{max}$

$x_{max} \sim \underbrace{(F(x))^n}_{\Psi}$

$\psi = \Psi'(x) = n(F(x))^{n-1} \cdot F'(x) = n \cdot \left(\frac{x}{\Theta}\right)^{n-1} \cdot \frac{1}{\Theta} \quad [(0;+\infty)]$

$M x_{max} = \int_0^{\Theta} x n \left(\frac{x}{\Theta}\right)^{n-1} \frac{1}{\Theta} dx = \frac{n}{\Theta^n} \cdot \frac{\Theta^{n+1}}{n+1} = \frac{n}{n+1} \cdot \Theta$ смещ.

$\rightarrow \tilde{\Theta}_3' = \frac{n+1}{n} \tilde{\Theta}_3 = \frac{n+1}{n} \cdot x_{max}$

$M\tilde{\Theta}_3' = \frac{n+1}{n} M\tilde{\Theta}_3 = \Theta$ несмещ.

$D\tilde{\Theta}_3 = \left(\frac{n+1}{n}\right)^2 \cdot D x_{max} = \frac{n+1}{n} \cdot \frac{n\Theta}{(n+2)(n+1)^2} = \frac{\Theta^2}{n(n+2)} \xrightarrow[\forall \Theta]{n \to \infty} 0$

$M x_{max}^2 = \int_0^{\Theta} x^2 n \left(\frac{x}{\Theta}\right)^{n-1} \frac{1}{\Theta} dx = \frac{n}{\Theta^n} \frac{\Theta^{n+2}}{n+2} = \frac{\Theta^2 n}{(n+2)}$

$D x_{max} = \Theta^2 \left(\frac{n}{n+2} - \frac{n^2}{(n+1)^2}\right) = \frac{\Theta^2 n}{(n+2)(n+1)^2}$

$\tilde{\Theta}_3$ по опред

$\forall \Theta \in \Theta \quad \forall \varepsilon > 0 \quad P(|\tilde{\Theta}_3 - \Theta| \geq \varepsilon) \xrightarrow[n \to \infty]{} 0$

$P(|x_{max} - \Theta| \geq \varepsilon) = P(x_{max} \leq \Theta - \varepsilon) = \Phi(\Theta - \varepsilon) =$

$= (F(\Theta - \varepsilon))^n = \left(\frac{\Theta - \varepsilon}{\Theta}\right)^n \longrightarrow 0$

$\frac{n+1}{n} M\tilde{\Theta}_3 -$ $\quad M\tilde{\Theta}_3$ $\quad$ I способ

$\forall \varepsilon > 0 \quad P(|\tilde{\Theta}_3' - \Theta| > \varepsilon) \xrightarrow{n \to \infty} 0$

$P(|\tilde{\Theta}_3' - M\tilde{\Theta}_3'| > \varepsilon) \leq \frac{D\tilde{\Theta}_3'}{\varepsilon^2} = \frac{\Theta^2}{(n+2)n\varepsilon^2} \to 0 \Rightarrow \tilde{\Theta}_3'$ - сост.

$$\tilde{\theta}_4 = x_1 + \frac{1}{n-1}\sum_{i=2}^{n} x_i$$

$$M\tilde{\theta}_4 = Mx_1 + \frac{1}{n-1}\cdot\sum_{i=2}^{n} Mx_i = M\xi + \frac{n-1}{n-1}M\xi =$$

$$= \frac{\theta}{2} + \frac{\theta}{2} = \theta$$

$$D\tilde{\theta}_4 = Dx_1 + \frac{1}{(n-1)^2}\sum_{i=2}^{n} Dx_i = \frac{\theta^2}{12} + \frac{n-1}{(n-1)^2}\cdot\frac{\theta^2}{12} \not\to 0$$

дост. усл. не вып.

по опред.

$$x_1 + \frac{1}{n-1}\sum_{i=2}^{n} x_i \xrightarrow{P} \xi + \frac{\theta}{2}$$ не явл. состоятельной

$\tilde{\theta}_3'$ по опред. <u>II способ</u>

$$\tilde{\theta}_3' = \frac{n+1}{n}\tilde{\theta}_3$$

$$\frac{n+1}{n}X_{max} - \theta = \frac{n+1}{n}(X_{max} - \theta) + \frac{\theta}{n}$$

$$\left|\frac{n+1}{n}X_{max} - \theta\right| = \left|\frac{n+1}{n}(X_{max} - \theta) + \frac{\theta}{n}\right| \le$$

$$\le \frac{n+1}{n}\left|X_{max} - \theta\right| + \frac{\theta}{n} = \frac{n+1}{n}\cdot(\theta - X_{max}) + \frac{\theta}{n}$$

$$\frac{n+1}{n}(\theta - X_{max}) + \frac{\theta}{n} > \varepsilon$$

$$\exists N: \forall n \ge N \hookrightarrow \frac{\theta}{n} < \frac{\varepsilon}{2} \Rightarrow \frac{n+1}{n}(\theta - X_{max}) > \frac{\varepsilon}{2}$$

$\Rightarrow$ для больших n $\theta - X_{max} > \frac{n}{n+1}\cdot\frac{\varepsilon}{2} \ge \frac{\varepsilon}{4}$

$$\Rightarrow P\left(\left|\tilde{\theta}_3' - \theta\right| > \varepsilon\right) \le P\left(\theta - X_{max} > \varepsilon/4\right)$$

$$P\left(X_{max} < \theta - \frac{\varepsilon}{4}\right) = \left(\frac{\theta - \varepsilon/4}{\theta}\right)^n \xrightarrow[n\to\infty]{} 0$$

$$\hat{\theta}_3 = \frac{n+1}{n} X_{max}$$

$$\tilde{\theta}_1 = 2\bar{X}$$

$$\tilde{\theta}_3' = \frac{\theta^2}{(n+2)n}$$

$$\frac{1}{3n} \not> \frac{1}{n(n+2)}$$

$$n^2+2n > 3n$$

$$n > 1$$

$\Rightarrow$ $\tilde{\theta}_3'$ более эфф., чем $\tilde{\theta}_1$